Week 2 | CLO 2-B | File 2 / 7

2-1 Organizing Data

تنظيم البيانات — التوزيعات التكرارية: الوصفية والمبوّبة وغير المبوّبة، والتكرار النسبي والمتجمع

Raw data (data in their original form) are too large to work with directly. Dr. Amani's motivating case is road accidents in Oman in 2024, with variables such as date, governorate, severity and day or night. The data are first **organised** into a frequency distribution (2-1), then **presented** with graphs (2-2, 2-3).

DEFINITIONS BOTH · CLO2-B

A **frequency distribution** is the organization of raw data in table form, using **classes** and **frequencies**. A **class** is a category or interval in which each data value belongs to exactly one class (mutually exclusive). The **frequency** f is the number of data values that fall into that class.

TYPE	DATA	USE WHEN	CLASSES LOOK LIKE
Categorical وصفي	qualitative (nominal or ordinal)	the data are categories	A, B, O, AB
Grouped مبوّب	quantitative	the range is **large**: each class is an interval wider than one unit	100–104, 105–109, …
Ungrouped غير مبوّب	quantitative	the range is **small**: each class is a single value	5, 6, 7, 8, 9, 10

1 · Categorical frequency distribution BOTH

STEPS

(1) Make a table: Class · Tally · Frequency · Percent. (2) Tally the data. (3) Count the tallies to get f. (4) Percent = `f/n × 100` . (5) Check: Σf = n and the percents total about 100%. Then **interpret** the table in a sentence.

EXAMPLE 2-1 · BLOOD TYPES OF 25 ARMY OFFICERS AHMED

CLASS	F	%
A	5	20
B	7	28
O	9	36
AB	4	16
Total	**25**	**100**

Type O is the most common (36%) and AB the least (16%).

TRUST IN INTERNET INFORMATION AMANI · EXERCISES 2-1

A = trusts everything, M = trusts most, H = trusts about half, S = trusts a small portion. Responses: M M M A H S M H M S M M M A M M M A M M H M H M M M M A M M M M M M.

CLASS	TALLY	F	%
A	||||	4	11.76
M	𝍸 𝍸 𝍸 𝍸 ||||	24	70.59
H	||||	4	11.76
S	||	2	5.88
Total		**34**	**99.99 ≈ 100**

The printed list has 34 responses. Interpretation: **about 71%** trust *most* of what they read online, while only **11.76%** trust *everything*.

The total is 99.99% because of rounding. Dr. Amani's note says the percent column should be "approximately 100", so 99.99 is accepted.

2 · Grouped frequency distribution BOTH

PROCEDURE (NUMBER OF CLASSES IS ALWAYS GIVEN IN THIS COURSE; USUALLY 5–20)

$$\text{width} = \frac{\text{range}}{\text{number of classes}} = \frac{H - L}{k} \rightarrow \textbf{always round UP to the next whole number } (4.2 \rightarrow 5,\ 4.8 \rightarrow 5,\ \textbf{4.0} \rightarrow \textbf{5})$$

(1) Number of classes k. (2) Width. (3) Starting point: the lowest value L (or a convenient number slightly less). (4) Lower limits: $LCL_1 = \min$, each next LCL = previous LCL + width; each $UCL = \text{next } LCL - 1$. (5) Boundaries: LCB = LCL − ½ unit, UCB = UCL + ½ unit. (6) Tally and count. (7) Relative frequency f/n. (8) Check Σf = n and ΣRF ≈ 1.

WHY A WHOLE-NUMBER WIDTH STILL GOES UP (4.0 → 5)

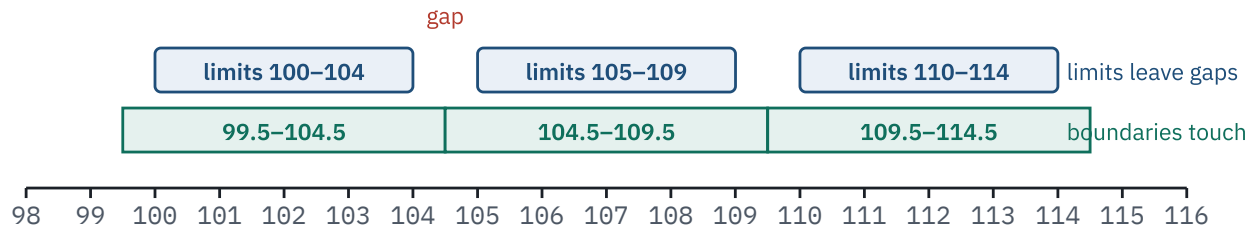
With whole-number data, k classes of width w cover values from L to L + k·w − 1, and that must reach H. So k·w must be **greater than** the range. Range 28 with 7 classes gives exactly 4, and 7 × 4 = 28 classes would stop at L + 27, one short of H. That is why the rule says round *up* even from a whole number.

BOUNDARIES: HALF THE SMALLEST UNIT AMANI

CLASS LIMITS	UNIT	½ UNIT	BOUNDARIES
15–20	1	0.5	14.5–20.5
2.2–10.3	0.1	0.05	2.15–10.35
4.26–7.20	0.01	0.005	4.255–7.205
12–24	1	0.5	**11.5–24.5**
10.24–15.02	0.01	0.005	**10.235–15.025**

IMPORTANT NOTES AMANI

Classes must be continuous: a class with no data still appears, with f = 0. There must be enough classes to cover all the data. All classes have the **same width**. The width can be read from a table three ways: **UCL − LCL + 1**; the next LCL − this LCL; the next LCB − this LCB (= UCB − LCB).



Example 2-2 (record temperatures). Class limits are the smallest and largest values a class can hold, so whole-number classes leave gaps (104 to 105). Boundaries subtract and add half a unit (0.5) so the classes meet, and the histogram bars can touch.

مثال 2-2: حدود الفئة (limits) أصغر وأكبر قيمة في الفئة، وبينها فجوات (من 104 إلى 105). الحدود الفعلية (boundaries) نطرح ونضيف نصف وحدة (0.5) فتتلاصق الفئات وتتلاصق أعمدة المدرج.

EXAMPLE 2-2 · RECORD HIGH TEMPERATURES OF 50 STATES, 7 CLASSES AHMED

112 100 127 120 134 118 105 110 109 112 · 110 118 117 116 118 122 114 114 105 109 · 107 112 114 115 118 117 118 122 106 110 · 116 108 110 121 113 120 119 111 104 111 · 120 113 120 117 105 110 118 112 114 114

R = 134 − 100 = 34; width = 34/7 = 4.86 → **5**. $LCL_1 = 100$, $UCL_1 = 100 + 5 - 1 = 104$; $LCB_1 = 99.5$, $UCB_1 = 104.5$.

LIMITS	BOUNDARIES	F	RF	CF	CRF
100–104	99.5–104.5	2	0.04	2	0.04
105–109	104.5–109.5	8	0.16	10	0.20
110–114	109.5–114.5	18	0.36	28	0.56
115–119	114.5–119.5	13	0.26	41	0.82
120–124	119.5–124.5	7	0.14	48	0.96
125–129	124.5–129.5	1	0.02	49	0.98
130–134	129.5–134.5	1	0.02	50	1.00

Most states' record highs lie between 110 and 119 °F (31 of 50 = 62%).

AGES OF DRIVERS, 7 CLASSES AMANI

21 21 21 23 23 23 23 24 25 25 26 26 26 26 27 27 28 28 28 28 28 29 29 29 29 29 30 30 30 30 31 31 31 31 31 31 32 34 35 35 35 36 36 37 37 37 38 38 39 41 42 42 42 43 43 43 43 44 44 44 44 45 45 45 46 47 48 48 48 49 49 49 49 50 50 51 51 65 72

R = 72 − 21 = 51; width = 51/7 = 7.29 → **8**; start at L = 21.

LIMITS	BOUNDARIES	F FROM THE PRINTED DATA	F IN DR. AMANI'S TABLE	RF (TABLE)
21–28	20.5–28.5	21	19	0.268
29–36	28.5–36.5	22	20	0.282
37–44	36.5–44.5	18	16	0.225
45–52	44.5–52.5	16	14	0.197
53–60	52.5–60.5	0	0	0
61–68	60.5–68.5	1	1	0.014
69–76	68.5–76.5	1	1	0.014
Total		**79**	**71**	**1**

The class limits, boundaries and the empty 53–60 class are the same in both versions; only the first four counts differ by 2 each. The 53–60 class stays in the table with f = 0 because classes must be continuous.

3 · Ungrouped frequency distribution AHMED

EXAMPLE 2-3 · HOURS OF SLEEP, 30 STUDENTS

8 6 6 8 5 7 7 8 7 6 6 7 9 7 7 · 6 8 10 6 7 6 7 8 7 7 8 7 8 9 8

Range = 10 − 5 = 5 (small), so one class per value.

LIMITS	BOUNDARIES	F	CF (≤ UCB)
5	4.5–5.5	1	1
6	5.5–6.5	7	8
7	6.5–7.5	11	19
8	7.5–8.5	8	27
9	8.5–9.5	2	29
10	9.5–10.5	1	30

7 hours is the most common (11 students); 19 of 30 sleep 7 hours or less.

4 · Relative, cumulative and cumulative relative frequency BOTH

FORMULAS

$$RF = \frac{f}{n} \qquad CF = f_1 + f_2 + \cdots \text{ (up to this class)} \qquad CRF = \frac{CF}{n}$$

RF can be a fraction, decimal or percent. CF counts the values **less than or equal to** a class's upper boundary. The first row ("less than the first LCB") is always **0**; the last CF is **n** and the last CRF is **1** (100%). Example: frequencies 5, 8, 12 give a third-class CF of 5 + 8 + 12 = 25.

DRIVERS, DR. AMANI'S TABLE (N = 71)

LESS THAN	CF	CRF
20.5	0	0
28.5	19	0.268
36.5	39	0.549
44.5	55	0.775
52.5	69	0.972
60.5	69	0.972
68.5	70	0.986
76.5	71	1.000

ACTIVE LEARNING · READING A FREQUENCY DISTRIBUTION AMANI

LIMITS	BOUNDARIES	F
10.0–19.9	9.95–19.95	3
20.0–29.9	19.95–29.95	7
30.0–39.9	29.95–39.95	12
40.0–49.9	39.95–49.95	9
50.0–59.9	49.95–59.95	6
60.0–69.9	59.95–69.95	3

(1) Class width? (2) Number of classes? (3) Range of the data set? (4) Which class has the highest frequency?

(1) **10**: 20.0 − 10.0, or 19.95 − 9.95. (Not 9.9: UCL − LCL + *one unit*, and the unit here is 0.1, gives 19.9 − 10.0 + 0.1 = 10.) (2) **6**. (3) The exact range needs the raw data. The table only bounds it: at most `69.95 − 9.95 = 60`, and the answer usually expected is **about 60**. (4) **30.0–39.9, f = 12**. Also n = 40.

الشرح بالعربي — بناء الجدول التكراري المبوّب

(1) عدد الفئات معطى دائمًا. (2) **الطول** = المدى ÷ عدد الفئات، ونقرّبه **لأعلى دائمًا**، حتى لو كان عددًا صحيحًا (4.0 تصبح 5)، وإلا فلن تدخل أكبر قيمة في آخر فئة. (3) أول حد أدنى = أصغر قيمة. (4) الحد الأدنى التالي = السابق + الطول، والحد الأعلى = الحد الأدنى التالي − 1. (5) **الحدود الفعلية**: نطرح ونضيف نصف أصغر وحدة (0.5 للأعداد الصحيحة، 0.05 لخانة عشرية، 0.005 لخانتين). (6) نفرز ونعدّ. (7) التكرار النسبي `f/n` ، والمتجمع نجمعه تراكميًا، أول قيمة فيه صفر وآخر قيمة n. (8) الفئة الفارغة **لا تُحذف**. انتبه: د. أماني كتبت أن الطول = (الحد الفعلي الأعلى − الأدنى) + 1، وهذا خطأ. الصحيح: الحد الأعلى − الحد الأدنى + 1، أو الحد الفعلي الأعلى − الحد الفعلي الأدنى.

Chapter 2 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

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